theraCUES™
SIGNALS FOR LIFE

RNA-seq (Eukaryotic Poly-A | Non-Model Organism - De Novo)

TC-S-BT-EPA-NMD

Applications

Strain variations or lack of an available nearest reference require creation of a de novo transcriptome assembly. De novo transcriptome assembly improves read mapping and annotation, especially for plant and fungal samples where reference variation limits suitability. Suitable for projects where an isolate is characterized for changes across treatments or developmental stages.

Pros

- Suitable for samples requiring reference genome creation
- High sensitivity for polyadenylated RNA
- Accurate expression quantification after assembly
- Suitable for evolutionary and ecological studies

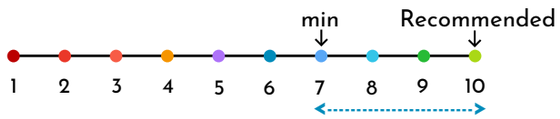
Notes

- Higher computational complexity than reference-based RNA-seq
- Not suitable for highly degraded RNA

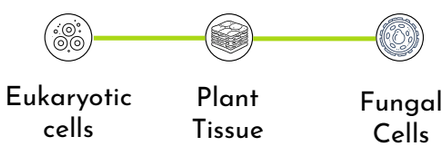
Input

Total-RNA Conc: 100ng - 500ng

RNA Integrity (RIN)



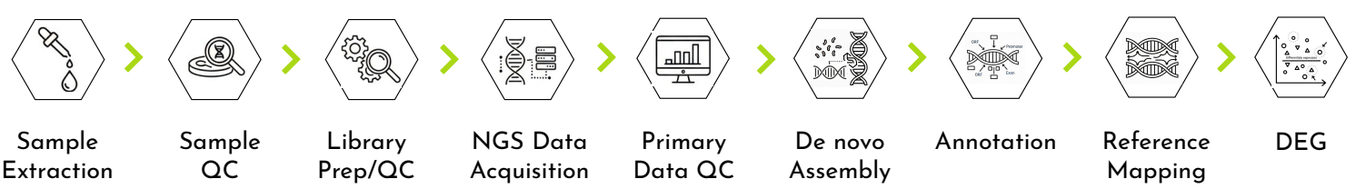
Sample Types



Platform

Illumina platforms | 150 x 2

Process Map



Standard Deliverables

Download Sample Report

- Raw & QC Trimmed data (.fastq files) and QC reports per sample
- Mapped data(.BAM) files sample and related QC reports
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology – Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathview maps*)
- *KEGG Pathview maps for pathways selected according to the interest of the study

References

- Sharma E et al. Front Mol Neurosci. 2025; PMID:40636521.
- Nilawar S, Chatterjee K. Biomacromolecules. 2022; doi:10.1021/acs.biomac.1c01235.
- Lalitha AV et al. Indian J Crit Care Med. 2024; PMID:39360202.